



Experiment 2

(Data Visualization)

Aim: Perform time series aggregation, apply filters on a given dataset, create line and area charts.

Theory:

Time aggregation:

Time aggregation is the aggregation of all data points for a single resource over a specified period (the granularity). Data aggregations in Resource Time Series reports are of the time aggregation type.

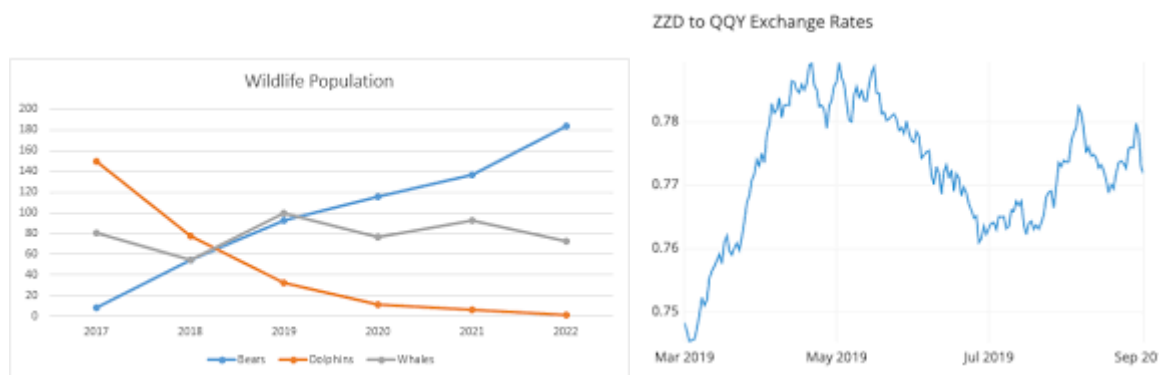
The result of the aggregation is one data point that reflects a statistical view of the collected and aggregated data points. For example, average, minimum, maximum, sum, or count. Typically, multiple aggregated data points are presented in a report for a given reporting period.

Data Filtering:

Data filtering is the process of choosing a smaller part of your data set and using that subset for viewing or analysis. Filtering is generally (but not always) temporary – the complete data set is kept, but only part of it is used for the calculation.

Line Charts:

A line chart displays information as a series of data points called 'markers' connected by straight line segments. It is a basic type of chart common in many fields.

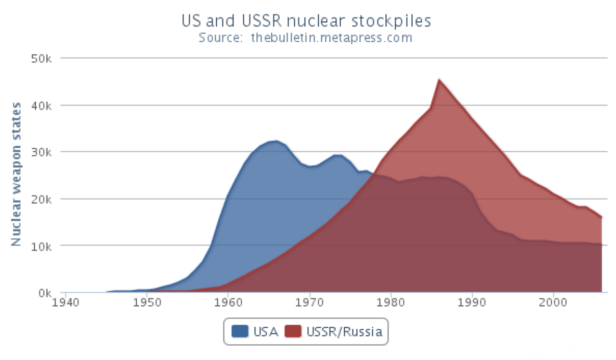


Area Charts:

An area chart is like a line chart in terms of how data values are plotted on the chart and connected using line segments. In an area chart, however, the area between the line segments and the x-axis is filled with color. It helps in comparison of multiple series, relationships between two series' and analysis of various trends.



Department of Computer Science and Engineering (Data Science)



Lab Assignments to complete in this session

Use the given dataset and perform the following tasks:

Dataset: 1

<https://raw.githubusercontent.com/nytimes/covid-19-data/master/us.csv>

Compatible Web Data Connector for easy importing:

<https://basic-csv-wdc.herokuapp.com/>

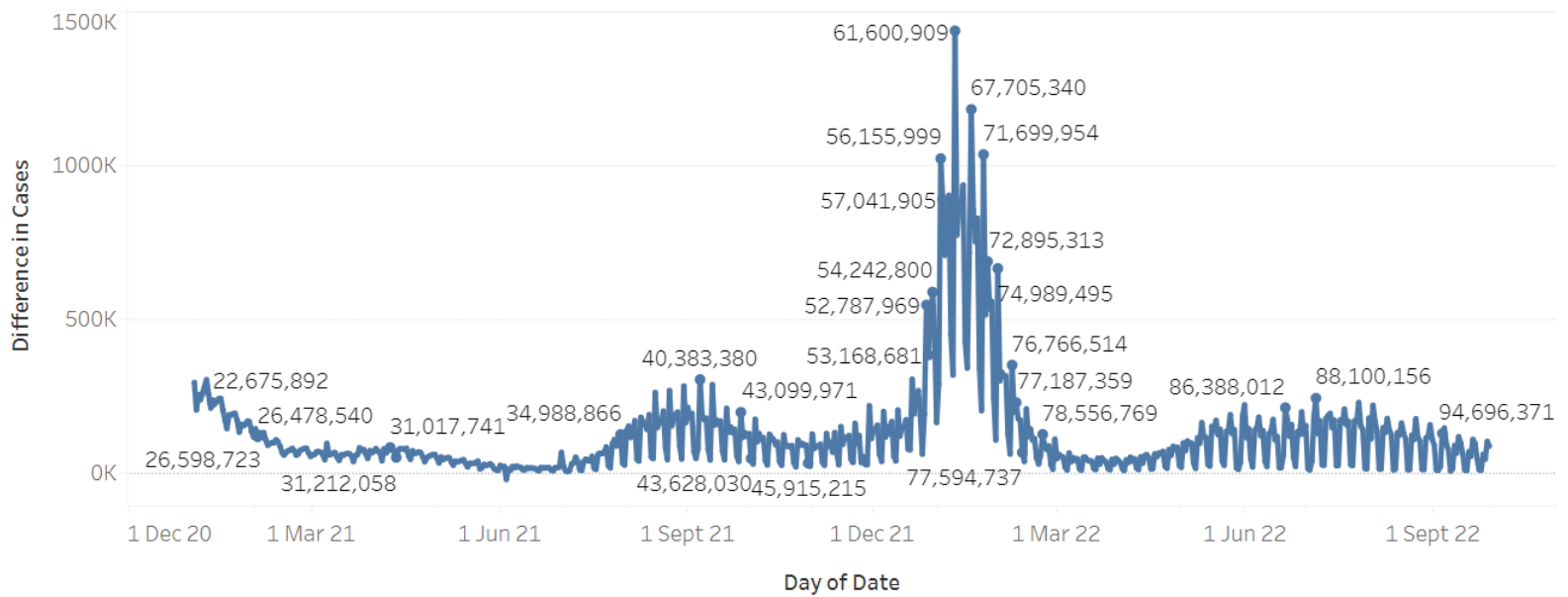
- Vaccination in the USA started in full swing in Early Jan 2021. Analyze the trend of Cases per day from Jan 2021 (Vaccines begun) to the current day and report if this policy was useful.
- The USA implemented a policy where Vaccinated Individuals could avoid wearing masks in public from June 2021. The UN wants to view the impact of making such a move in the long term as well as the short term.
- Analysis requires the creation of a calculated field in terms of Cases Per day and a Weekly moving average of cases per day.
- Justify why you have chosen a particular visualization for this dataset. Also discuss the details of your visualization.

Dataset: 2

Creditcard.csv from Kaggle

- Using appropriate visualization find out the dates when credit card fraud had taken place.
- Justify why you have chosen a particular visualization for this dataset. Also discuss the details of your visualization.

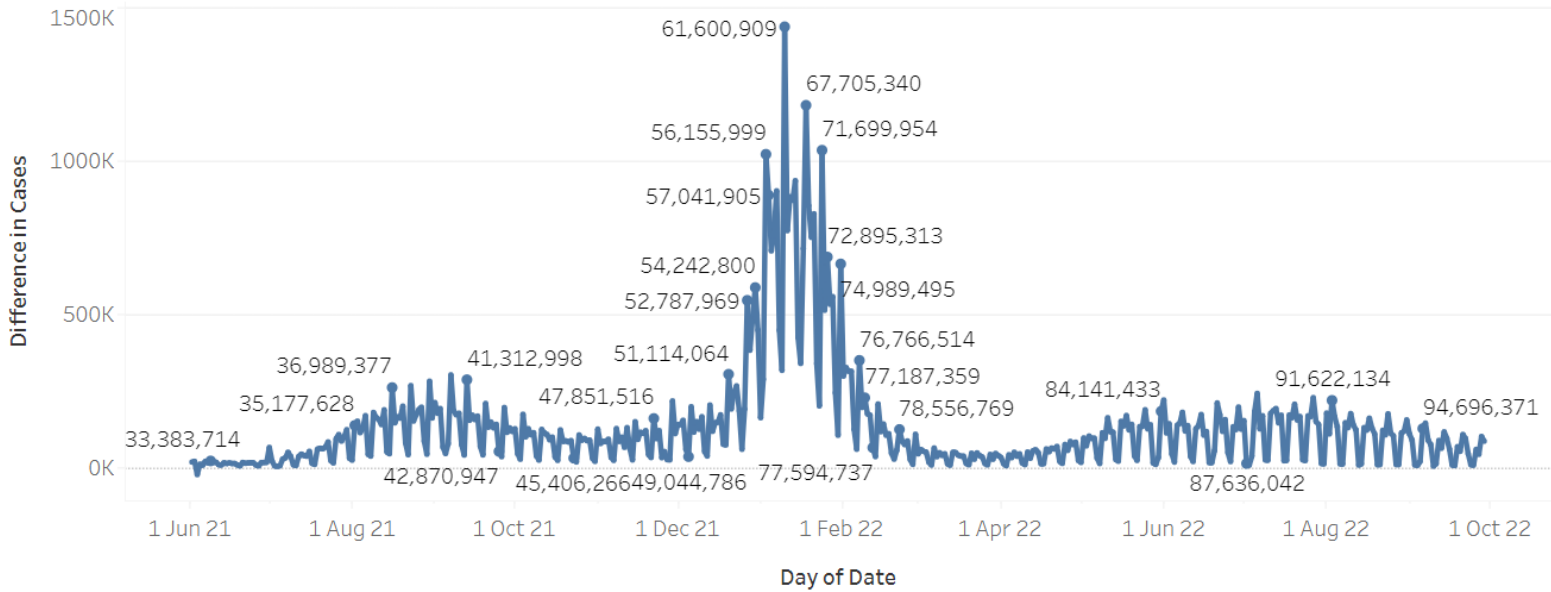
Q2)a)



Analysis: From the above graph it is seen that the number of cases per day decreased since January 2021, when vaccination was introduced. This tells us that the vaccination policy was successful. Although there was a spike in the number of covid cases in January 2022.

b) In long term:

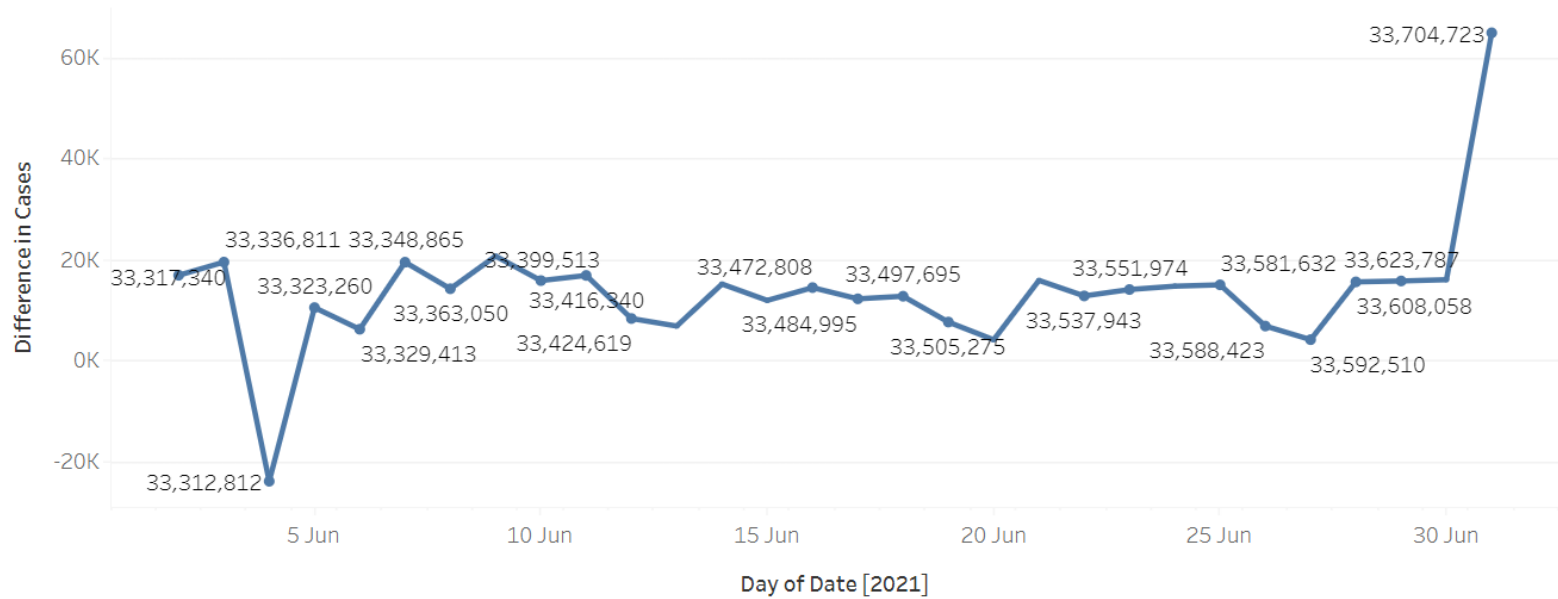
Sheet 2



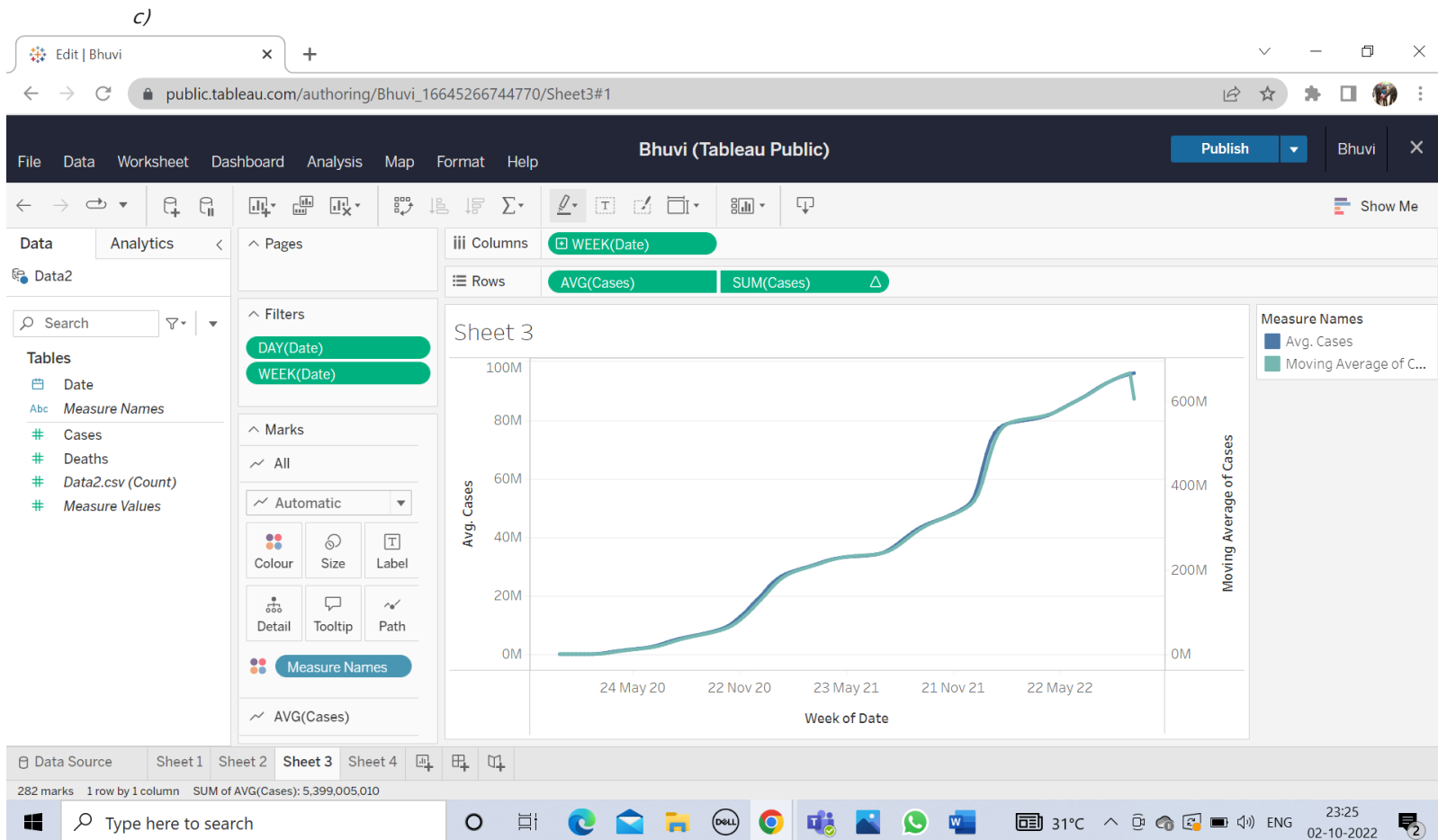
Analysis: The policy doesn't seemed to have had any meaningful decline in the number of cases in the long term, the decrease in cases in the long term seems to have been a result of the vaccination drive rather than the policy of removing mask compulsions.

In short term:

Sheet 2



Analysis: There has been an increase in the number of cases in June 2021 as wearing masks was not compulsory



Analysis: The moving average depicts a steady increase in the number of cases per day.

d) The bar plots are a great way to visualise the number of cases in each country as the height of the bar directly corresponds to the number of cases, making it easy to understand.

The line plot is a fantastic way to judge the trends in the number of cases over time. Plotting a bar graph in this case will make it messy and we might miss important trends in data if we increase the intervals of the bar graph in this case in favour of visual representation. Hence line plot is the best way to analyse trends over time. Same goes for analysing the moving average over a particular period of time.

Conclusion: Line plot has been used to show the variation in covid cases over a period of time.